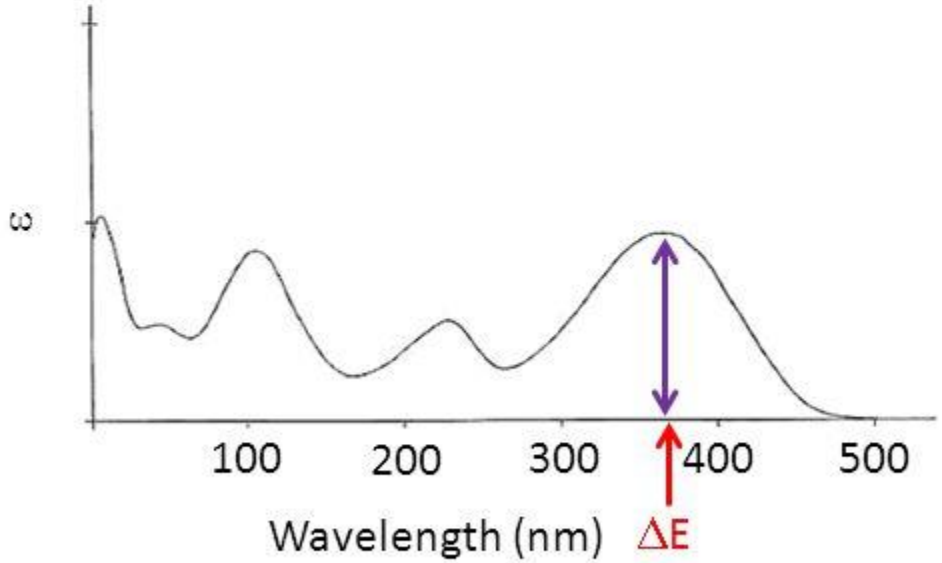
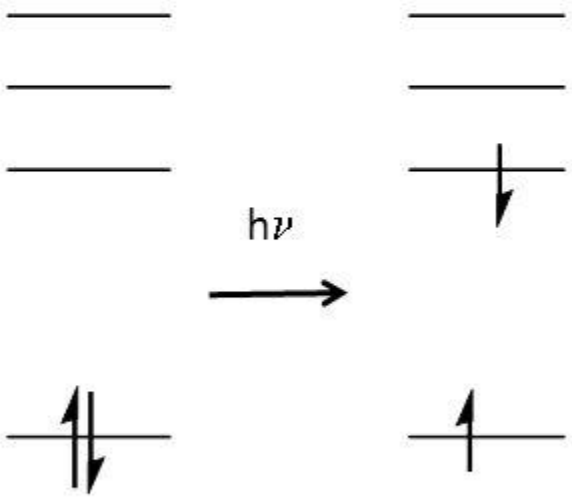
Electronic Transitions

Spectral Features:

- Number of transitions.
- Energy of the transitions.
- Intensity of the transitions.
- Shape of the transition.





Transition Probability

Transition Probability

Consider two non-degenerate stationary states *m* and *n* of a system with energy values E_m and E_n, where $E_m > E_n$. Hence, the frequency of the absorbed radiation is given by,

$$\nu_{mn} = \frac{E_m - E_n}{h}$$

Assume that the system is in the lower state *n* and exposed to a radiation of density $\rho(\nu)$.

The probability per unit time that it will absorb the radiation and will thereby raised to the upper state *m*, is proportional to the number of particles N_n in the state *n* and the density of radiation of frequency ν_{nm}. Hence,

$$\text{Rate of absorption} = B_{nm} N_n \rho \nu_{nm}$$

where B_{nm} is the proportionality constant and is known as **Einstein's coefficient of absorption**.

Transition Probability

The probability of return from *m* to *n* consists of two parts:
(i) spontaneous (independent of radiation density) and
(ii) Induced (proportional to radiation density)

If N_m be the number of particles in the upper state *m* at any time *t*, then

$$\text{Rate of emission} = A_{mn}N_m + B_{mn}N_m\rho\nu_{mn}$$

where A_{mn} known as **Einstein's coefficient of spontaneous emission** and B_{mn} as **Einstein's coefficient of induced emission**.

At equilibrium, the two rates will be equal.

$$B_{nm}N_n\rho\nu_{nm} = A_{mn}N_m + B_{mn}N_m\rho\nu_{mn}$$

On rearrangement,

$$\frac{N_n}{N_m} = \frac{A_{mn} + B_{mn}\rho\nu_{mn}}{B_{nm}\rho\nu_{nm}}$$

Boltzmann Distribution Law and Population in Energy States

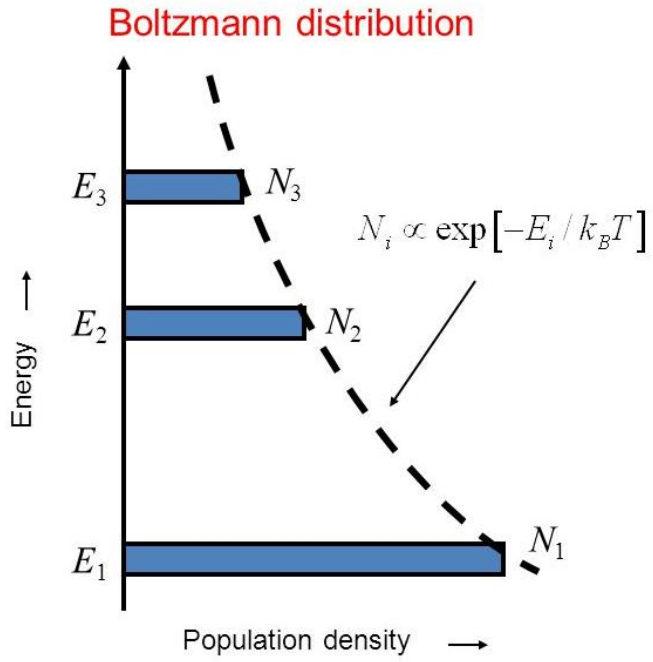
The **population in any energy state** is governed by the **Boltzmann distribution law**, i.e., at equilibrium

$$N_x = N_0 e^{-E_x/kT}$$

where, N_x is the number of particles in a given state x with energy E_x. N_0 is the total number of particles in the ground energy level, T is the absolute temperature, k is the Boltzmann constant = 1.38044 x 10^{-23} JK^{-1}.

If the number of molecules in the state with energy E_2 is N_2, and the number in the state with energy E_1 is N_1 then Boltzmann distribution law states that at equilibrium

$$\frac{N_1}{N_2} = \frac{e^{-E_1/kT}}{e^{-E_2/kT}} = e^{\Delta E/kT}$$



The **population difference** between the two states may be deduced from the population ratio expression as follows:

$$\frac{N_1 - N_2}{N_1 + N_2} = \frac{e^{\Delta E/kT} - 1}{e^{\Delta E/kT} + 1}$$

$$\frac{N_1 - N_2}{N_1 + N_2} = \frac{(N_1/N_2) - 1}{(N_1/N_2) + 1} = \frac{e^{\Delta E/kT} - 1}{e^{\Delta E/kT} + 1}$$

Since $N_1 + N_2 = N$ and $e^x = 1 + x + \frac{x^2}{2!} + \ldots.$ and $\frac{\Delta E}{kT} \ll 1$

Therefore $$\Delta n = N_1 - N_2 = \frac{N\Delta E}{2kT}$$

Thus, the **population difference** Δn relates <u>directly</u> to the **total population**, **the energy difference between the two states** and <u>inversely</u> to the **absolute temperature**.

i.e., the **net intensity of the absorption line** depends on the **total population in the ground state**, **the energy difference between the two states**, **absolute temperature** and **population difference between the two states at equilibrium** (provided intensity of the radiation is kept constant)

Average Lifetime of the Excited State

The transition probability equals the inverse of the average lifetime of the excited state (τ).

$$A_{mn} = {}^{1}/_{\tau}$$

If τ is small A_{mn} will be large and intensifies the signal.

Concentration and Path Length of Sample

The **net intensity of absorption** is related to the concentration C of the sample and the path length l by **Beer-Lambert law**.

$$\boxed{T = \frac{I_t}{I_0} = e^{-\varepsilon' Cl}}$$

$$\boxed{A = log\frac{I_t}{I_0} = \frac{\varepsilon' Cl}{2.303} = \varepsilon Cl}$$

$$\boxed{\varepsilon' = 2.303\varepsilon}$$

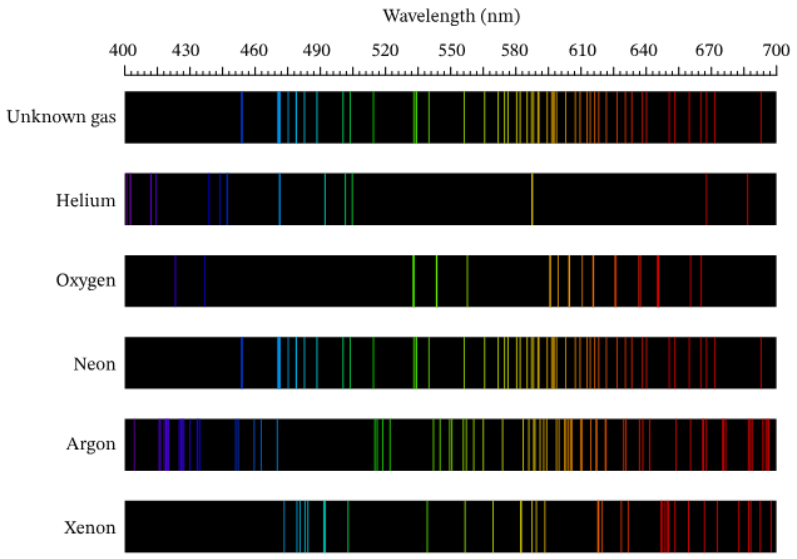
where, I_t is the intensity of the transmitted radiation, I_0 the intensity of incident radiation, A is the absorbance, C the concentration of the absorbing species, l path length and $\varepsilon'$is the molar extinction or molar absorbance or molar decadic coefficient (ε).

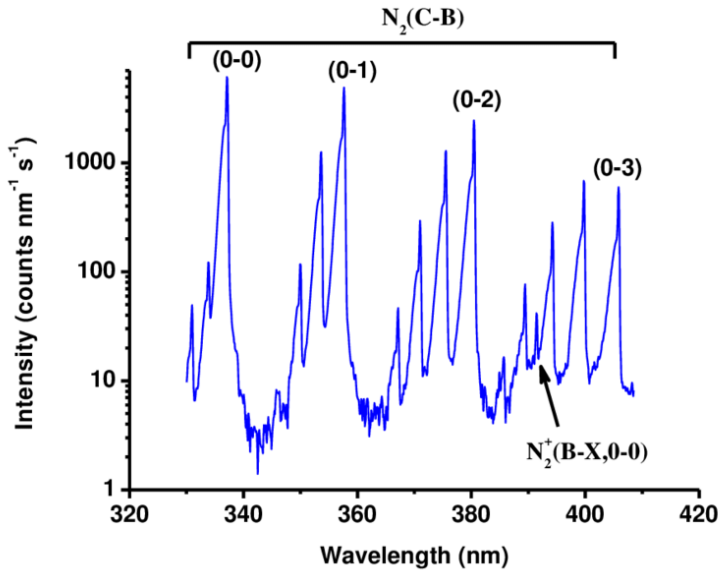
Atomic Structure and Atomic Spectra

Interpretation of atomic and molecular spectra has a vital role in quantum mechanics.

The light from an electrostatic discharge through a tube containing a monatomic gas at low pressure shows a series of distinct lines (groups) characteristic to the gas – **line spectrum of the gas**

Since atoms are the source of these lines, so the line spectra are known as **atomic spectra**.





Molecules give **band spectra** (regular arrangement of very closely spaced lines).

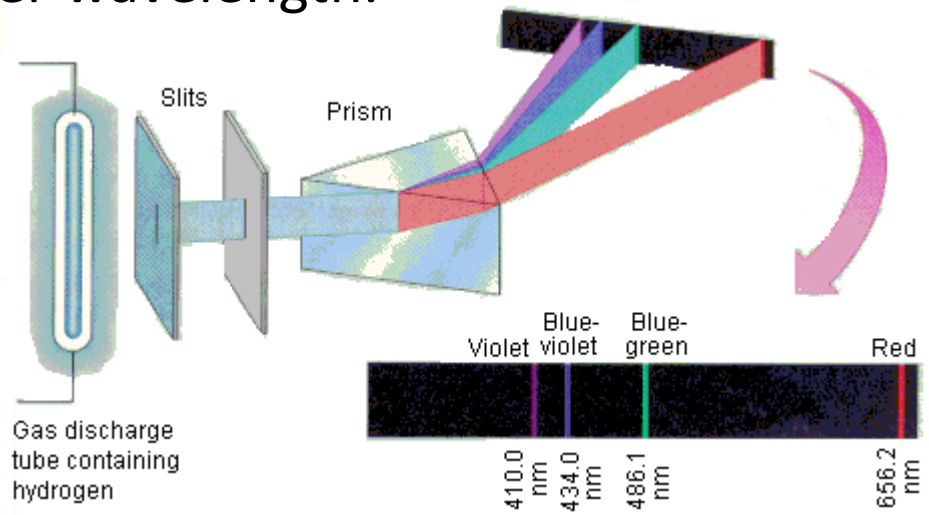
The band spectrum disappears when the emitting gas is subjected to a high voltage discharge at which the molecule dissociates to atoms.

Hence, the origin of band spectrum is molecules, and it is called **molecular spectrum**.

The main features

- The atomic spectra have linear structure.
  - This enables to describe the spectrum of each element by specific values of wavelength λ or frequency ν.

- Separate groups of series of lines in atomic spectra obey definite series laws.
  - These laws are particularly prominent in the spectrum of simplest atom, for example: hydrogen.
  - The separation between succeeding lines in a series diminishes gradually a we move toward the shorter wavelength.



History

- 1859 – Kirchoff demonstrated that elements and compounds have characteristic spectra.

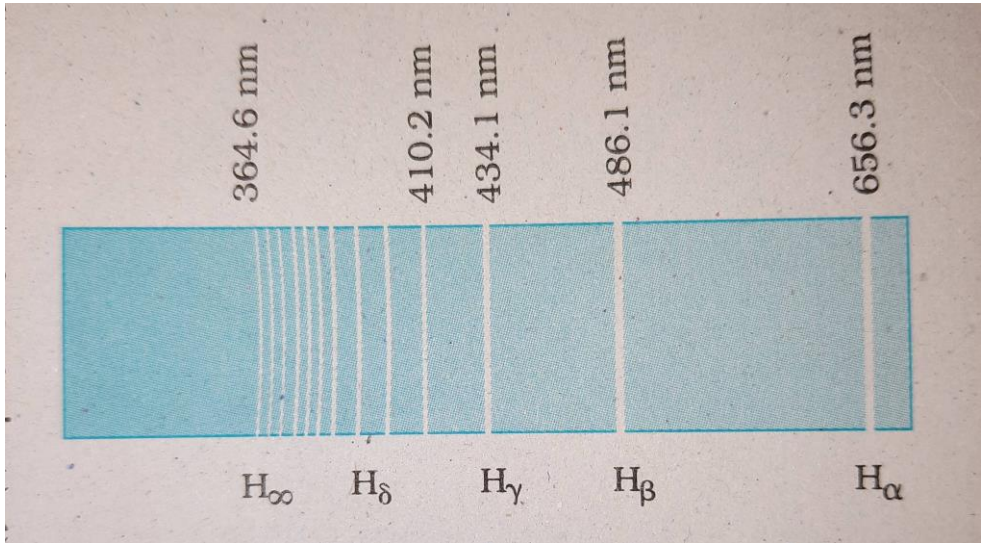
- 1861 – Kirchoff and Bunsen discovered a fourth alkali metal, cesium, while studying the spectra and later rubidium (the fifth alkali metal).

- 1885 – The first quantitative interpretation of atomic spectra began when Balmer demonstrated that the wavelengths of the nine lines of H-atom are expressed by the formula:

$$\lambda = \lambda_0 \frac{m^2}{m^2 - n^2}$$

where, λ_0 = 364.56 nm, n = 2 and m = 3,4,5,.....11. A spectrogram of this series of hydrogen spectrum is called the ***Balmer series***.

The lines in successive order are known as H_α, H_β, H_γ, H_δ ..... and so on.



- Rydberg (1854-1919) proposed an empirical formula from which the **wavelength of the Balmer lines could be calculated**.

$$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{m^2}\right), \qquad m = 3,4,5,....$$

The Rydberg constant, R = 1.0973731 x 10^{-3} Å

- Balmer suggested the existence of a series characterized by n = 3. This was actually discovered later on by Paschen in 1908. The Paschen series is expressed by:

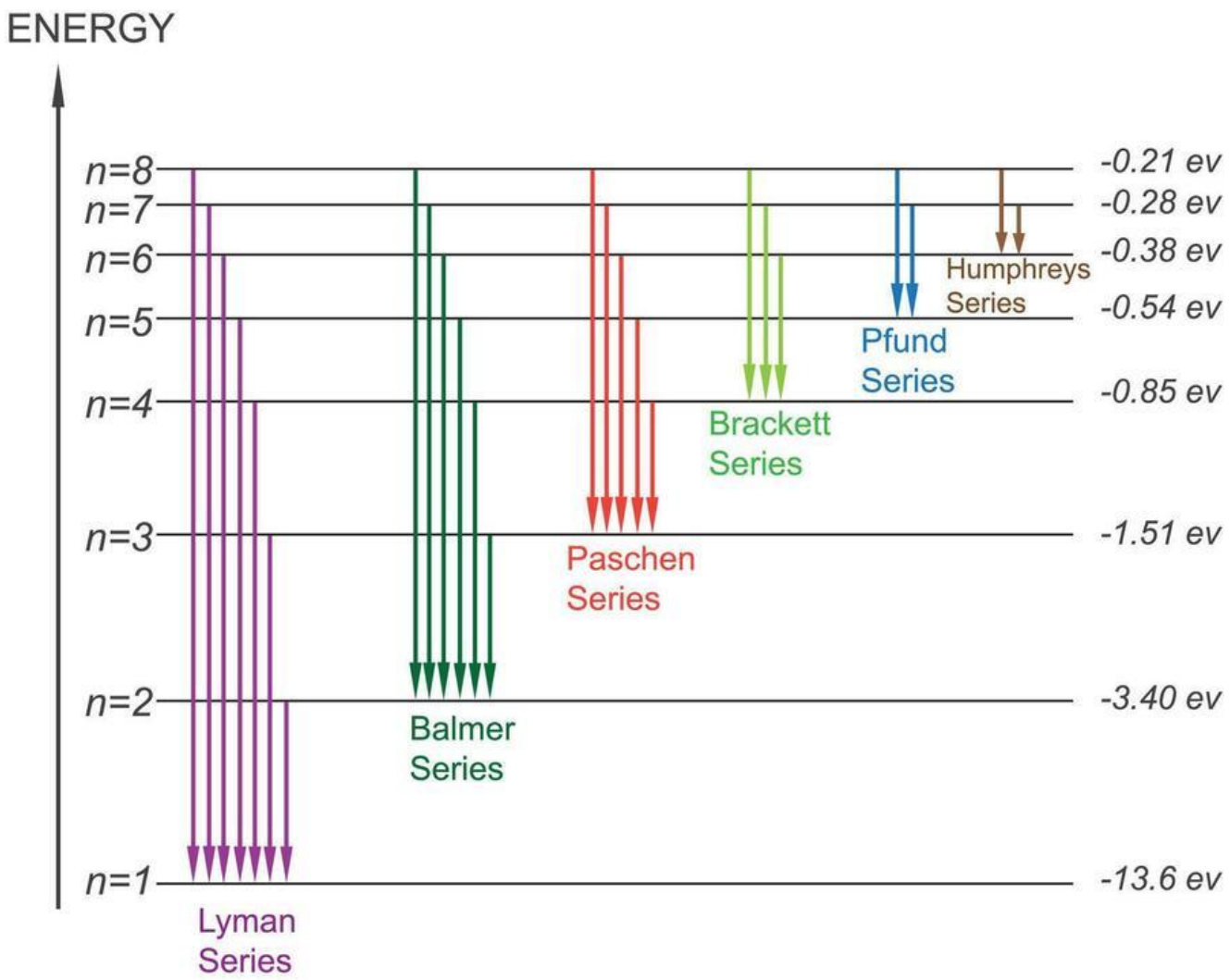
$$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{3^2} - \frac{1}{m^2}\right), \qquad m = 4,5,6,....$$

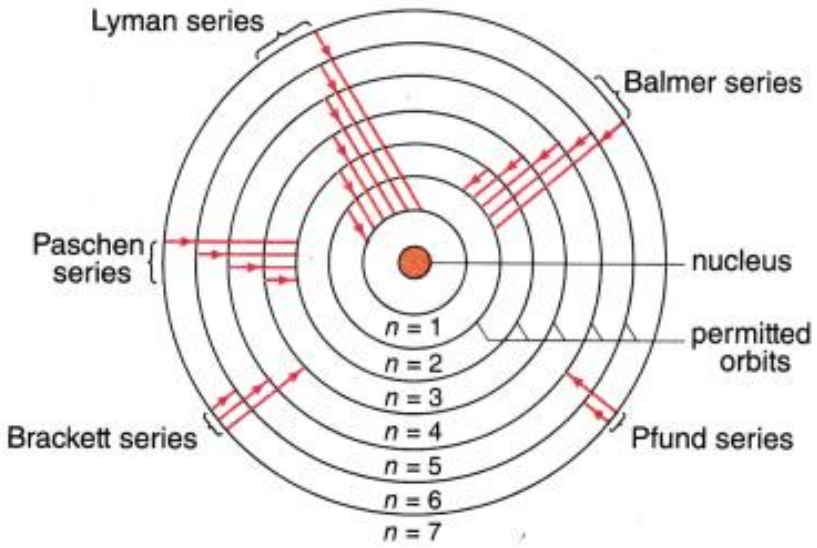
- Lyman discovered a series that obeys the formula:

$$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{m^2}\right), \qquad m = 2,3,4,....$$

The Balmer formula was taken as the basis for the theory of spectroscopy.

Line Spectrum Of Hydrogen atom





Ritz Combination Principle

If λ_1 and λ_2 are the wavelengths of two lines in the spectrum of an atom, then it may so happen that lines also occur at wavelengths $\frac{1}{\lambda_1} + \frac{1}{\lambda_2}$ or $\frac{1}{\lambda_1} - \frac{1}{\lambda_2}$.

For Balmer series of H-atom spectrum,

$$\frac{1}{\lambda_\alpha} = R_H\left(\frac{1}{2^2} - \frac{1}{3^2}\right), \quad \frac{1}{\lambda_\beta} = R_H\left(\frac{1}{2^2} - \frac{1}{4^2}\right) \quad \text{and} \quad \frac{1}{\lambda_\beta} - \frac{1}{\lambda_\alpha} = R_H\left(\frac{1}{3^2} - \frac{1}{4^2}\right)$$

The first line of Paschen series

- 1902 – Planck postulated that light is emitted or absorbed by an atom only in definite portions or quanta of magnitude hν.

$$\Delta E = h\nu$$

- 1911 – Bohr proposed that an atom has a set of stationary energy states and energy states of all atoms follow Planck's formula.
  - The transition from one state E_n to another state E_m is accompanied by emission or absorption of energy $\Delta E = |E_n - E_m|$.

These postulates led to the foundation of spectroscopic methods for obtaining quantitative information about the separation of energy states in atoms and molecules.

The actual energy states and allowed transitions between them are determined from the quantum mechanical selection rules.

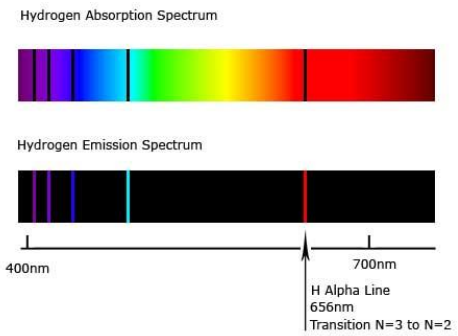
The spectra of ionized helium, beryllium and other elements with one electron exhibit series similar to those observed in the spectra of hydrogen atom.

The series of hydrogen-like He^{+}, Li^{2+}, Be^{3+}, etc. are expressed by the formula:

$$\bar{\nu} = \frac{1}{\lambda} = Z^2 R \left(\frac{1}{n^2} - \frac{1}{m^2} \right), \text{ m = n+1, n+2, n+3,....}$$

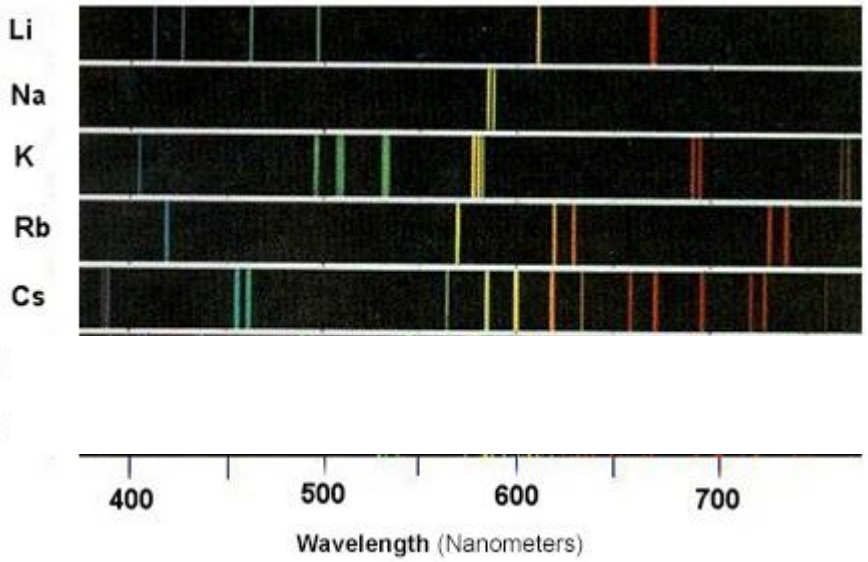
1912 – Franck and Hertz demonstrated the presence of excited states in an atom.

Spectra of Alkali Metals



The series of alkali metal atoms

Elements	Atomic Number	Electronic Configuration
Lithium	3	$1s^2\ 2s^1$
Sodium	11	$1s^2\ 2s^2\ 2p^6\ 3s^1$
Potassium	19	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$
Rubidium	37	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 5s^1$
Caesium	55	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 5s^2\ 5p^6\ 6s^1$



This suggests that an alkali atom consists of one or more closed shell of electron in its ground state and a single valance electron in a new shell in ns orbit.

The series regularities like those of hydrogen atoms are also observed in the spectra of the alkali metals.